

1. Verify the signature

Step 1: The device runs a correctly signed program. Upgrade the correctly signed firmware.

-Verify signature successfully:

```
head check ok.
checkUPgradeSD checkUPgradeHeade suc.headLen:768 ret:10240 !
lfdebug func:checkUPgradeSD line:2197
lfdebug func:checkUPgradeSD line:2199
[DevCapture.cpp][936]938 1 IPC 1 IPC

func:OSD_SetTitle line:989
func:OSD_SetTitle line:1048: pCaptureDev->vstd:0,video_size:5
func:OSD_SetTitle line:1078 string:IPC
rect.left=0,rect.top=6,endX=368,endY=17,size.w=352,size.h=288,width=23
func:OSD_SetTitle line:989
func:OSD_SetTitle line:1048: pCaptureDev->vstd:0,video_size:17
1105[OSD_SetTitle]: time_format:0,time_seprator:0,time_lang:1
rtstream: <rts_av_wait, 68>[rts_av_do_cmd]lock depth is 2
rtstream: <rts_av_wait, 68>[rts_av_get_osdi]lock depth is 2
rtstream: <rts_av_wait, 68>[rts_av_do_cmd]lock depth is 2
rtstream: <rts_av_wait, 68>[rts_av_get_osdi]lock depth is 2
update_osdi_timedate, query osdi attr success
Signature verified successfully!
07:54:35[TRACE] SDRecord.cpp:checkUPgradeSD:2240: veriy pak suc.
LIBDVR : SystemShell [rm -rf /var/upgrade]
LIBDVR : SystemShell [rm -rf /var/upgrade] success
LIBDVR : SystemShell [mkdir -p /var/upgrade]
LIBDVR : SystemShell [mkdir -p /var/upgrade] success
07:54:35[TRACE] Upgrader.cpp:notify:759: CUpgrader::notify Preparing, 0%,
07:54:35[INFO] AppEvent.cpp:notify:372: CAppEventManager-----appEventPulse.
07:54:35[TRACE] RtspThread.cpp:Start:110: CRtspSrv:CRtspSrv Start Successful,<0> is listennig on <554>
```

Step 2: The device runs the program with the correct signature. Upgrade the firmware with the wrong signature.

-Failed to verify signature:

```
func:OSD_SetTitle line:989
func:OSD_SetTitle line:1048: pCaptureDev->vstd:0,video_size:5
func:OSD_SetTitle line:1078 string:IPC
rect.left=0,rect.top=6,endX=368,endY=17,size.w=352,size.h=208,width=23
func:OSD_SetTitle line:989
func:OSD_SetTitle line:1048: pCaptureDev->vstd:0,video_size:17
1105[OSD_SetTitle]: time format:0,time_seperator:0,time_lang:1
rtstream: <rts_av_wait, 68>[rts_av_do_cmddlock depth is 2
rtstream: <rts_av_wait, 68>[rts_av_get_osdijlock depth is 2
rtstream: <rts_av_wait, 68>[rts_av_do_cmddlock depth is 2
rtstream: <rts_av_wait, 68>[rts_av_get_osdijlock depth is 2
update_osdi_timedate, query osdi attr success
Signature verification failed.
very pak failed.
signature:de85d9644de79b590cc69f82d87d1albe0c0ec9b0e8a90bda0a07a013e0a2b0ec7e73c9154eeb28f67d2ef65aeab9ca7dfce68f2020ff6dcbe79ff69ba3b5b1lec
a96a7aa6c44265724bb02b157d079013149da0ec36a25b40ab3ce5716fcf3b81ed764504dec7fe3776de7d423baa470d2933c50959de02ef36ee64ad90dba293c671177fd883b1e4
bed0835a22facad4ac6bca95ef2743e3c077d97b50d9b3a7b78795b5a9ea4a8621c85dfc77817d9da02d954efb3ca2354c7669a391314e15d9c6201eba9390ac2572af765a675
LIBDVR : SystemShell [rm -rf /var/upgrade] success
LIBDVR : SystemShell [mkdir -p /var/upgrade]
LIBDVR : SystemShell [mkdir -p /var/upgrade] success
07:54:35[TRACE] Upgrader.cpp:notify:759: CUpgrader::notify Preparing, 0%,
07:54:35[INFO] AppEvent.cpp:notify:372: CAppEventManager-----appEventPulse.
07:54:35[TRACE] RtspThread.cpp:Start:110: CRtspSrv:CRtspSrv Start Successful,<0> is listenng on <554>
```

Step 3: The device runs a properly signed program. Upgrade the unsigned firmware.

-Failed to verify signature:

[illegible]

2. Verify version

A: Upgrade from a lower version to a higher version

Step 1: The current version is lower, select a higher version, and the upgrade verification passes.

```
/var/run/wpa_supplicant: No such file or directory
Failed to connect to non-global ctrl_ifname: eth2
08:50:35|[TRACE] ZbarQRcodeScan.cpp:CheckWifiConnec

camera_set_led_status red=0, blue=4
camera_set_led_status red=0, blue=4
open VoicePlay fail! error [/var/voice/v_wifi_conne
/var/run/wpa_supplicant: No such file or directory
Failed to connect to non-global ctrl_ifname: eth2
Header Check success !
08:50:38|[TRACE] ZbarQRcodeScan.cpp:CheckWifiConnec
```

Step 2: Start Upgrading.

```
DAEMON : FILE -> common/flash.c, LINE -> 77: umount Custom Failed err[22]: Invalid argument
DAEMON : FILE -> common/flash.c, LINE -> 81: umount Custom Failed err[22]: Invalid argument
DAEMON : FILE -> common/flash.c, LINE -> 85: umount Web Failed err[2]: No such file or director
FlashInit: mtd0,start:0,size:50000,esize:10000 name:uboot"
FlashInit: mtd1,start:50000,size:2f0000,esize:10000 name:romfs"
FlashInit: mtd2,start:340000,size:430000,esize:10000 name:USER"
FlashInit: mtd3,start:770000,size:40000,esize:10000 name:CUSTOM"
FlashInit: mtd4,start:7b0000,size:50000,esize:10000 name:MTD"
FlashInit: mtd5,start:800000,size:800000,esize:10000 name:ALL"
partnum:0 partname:<userfs.cramfs> buflen:<0x42b000> partLen:<0x430000> addr:<0x340000> size:<0
offset=0x0, g_mtd_saddr=0x340000, g_mtd_esize=0x10000
start:0, length:10000
start:10000, length:10000
start:20000, length:10000
start:30000, length:10000
start:40000, length:10000
start:50000, length:10000
start:60000, length:10000
start:70000, length:10000
start:80000, length:10000
start:90000, length:10000
start:a0000, length:10000
start:b0000, length:10000
start:c0000, length:10000
start:d0000, length:10000
```

Step 3: The upgrade is successful and the device restarts.

```

partnum:2 partname:<romfs.cramfs> buflen:<0x2ea000> partLen:<0x2f0000> addr:<0x50000> size:<0x2f0000>
offset=0x0, g_mtd_saddr=0x50000, g_mtd_esize=0x10000
partnum:3 partname:<u-boot.bin> buflen:<0x3d8d0> partLen:<0x40000> addr:<0x0> size:<0x40000>
offset=0x0, g_mtd_saddr=0x0, g_mtd_esize=0x10000
partnum:4 partname:<u-boot.env.bin> buflen:<0x10000> partLen:<0x10000> addr:<0x40000> size:<0x10000>
offset=0x40000, g_mtd_saddr=0x0, g_mtd_esize=0x10000
start:40000, length:10000
[daemon-W] Daemon_UpgradeByNet.c:UpGradeByLocal_Pthread:1697 :OTA upgrade sucesed.
The system is going down NOW!
Sent SIGTERM to all processes
Requesting system rebootesses
[ 260.378846] fh_axi_dma_adap: Use nor flash.
ROM: Init DDR..Training done.
ROM: Ok

```

B: Upgrade from a higher version to a lower version

Step 1: The current version is higher , select a lowerversion, and the upgrade verification fails.

```

func:OnOTABegin line:4622
/var/run/wpa_supplicant: No such file or directory
Failed to connect to non-global ctrl_ifname: eth2 error: No such file or directory
09:06:01|[TRACE] ZbarQRcodeScan.cpp:CheckWifiConnectStatus:1403: Status:<wpa_cli

/var/run/wpa_supplicant: No such file or directory
Failed to connect to non-global ctrl_ifname: eth2 error: No such file or directory
Header invalid. new version: 001, old version: 004.
Header is invalid!
09:06:04|[TRACE] Record.cpp:Stop:2048: CRecordManager::Stop()>>>>>>>>

CZbarScan::StopSnapShot has stopped.
[Packet.cpp][1029]Packet usage : 129K / 8191K, 2%

09:06:06|[TRACE] Packet.cpp:PrintfPacketUsage:1052:
-----[10-15 09:06:06]

```

Step 2: Upgrade failed, exit upgrade, restart the device.

```

umount failed
09:06:07|[ERROR] Main.cpp:Reboot:919: OnReboot[ 113.582954] fh_wdt: [wdt] set topval: 4
LIBDVR : Sys Will Reboot
WDTInit Has Called
WatchdogClear success
mbedtls_ssl_conf_alpn_protocols absent
ssl_srv.c:2168 got ciphersuites in common, but none of them usable
ROM: Use nor flash.
ROM: Init DDR..Training done.
ROM: Ok

U-Boot 2010.06 (Jul 25 2023 - 13:50:28)

DRAM: 256 MiB

```